

WEEK – 6

Strings CO1

Write a program to find the longest Substring without Repeating Characters

Input: abcabcbb output:3 string: abc

Input: pwwkew output:3 string: wke

Note: pwke is not a substring , it is a subsequence

Steps For Execution:

Step 1: Read the string "abcabcbb".

Step 2: Start from index 0 and find "abc".

Step 3: The next character 'a' repeats, so stop.

Step 4: maxLength = 3 and longest = "abc".

Step 5: Check the remaining starting positions.

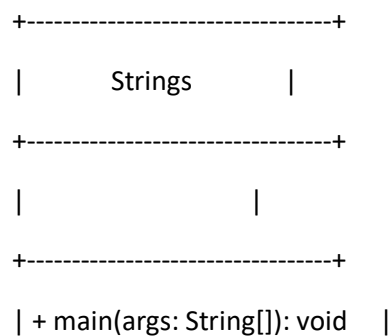
Step 6: No substring longer than 3 is found.

Step 7: Display the result.

Algorithm:

1. Start.
2. Read the input string s.
3. Initialize maxLength = 0 and longest = "".
4. Select each character as the starting position using an outer loop.
5. Initialize temp = "".
6. Check the characters using an inner loop.
7. If a character already exists in temp, stop the inner loop.
8. Otherwise, add the character to temp.
9. If temp.length() > maxLength, update maxLength and longest.
10. Repeat until all starting positions are checked.
11. Display maxLength and longest.
12. Stop.

UML Class Diagram:



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Code:

```
package javacore;

import java.util.Scanner;

public class Strings {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a string: ");
        String s = sc.next();

        int maxLength = 0;
        String longest = "";

        for (int i = 0; i < s.length(); i++) {
            String temp = "";

            for (int j = i; j < s.length(); j++) {
                char ch = s.charAt(j);

                if (temp.indexOf(ch) != -1) {
                    break;
                }

                temp = temp + ch;

                if (temp.length() > maxLength) {
                    maxLength = temp.length();
                    longest = temp;
                }
            }
        }

        System.out.println("Length: " + maxLength);
        System.out.println("String: " + longest);

        sc.close();
    }
}
```

Output:

```
Enter a string: abc
Length: 3
String: abc
```

